Single Linked List – Memory Allocation

We have seen:

```
Node *newNode = (Node *)malloc(sizeof(Node));
```

→ *The size of newNode will be equal to the size of a Node structure and newNode is declared as a pointer to Node structure.*

→ *The size of a pointer (newNode) is typically 8 bytes on most modern 64 – bit systems, regardless of the size of the structure it points to.*

→ *The size of the Node structure itself is determined by the sum of the sizes of its members*:

→ *data(an integer) is typically 4 bytes.*

→ *next(a pointer to another Node) is typically 8 bytes.*

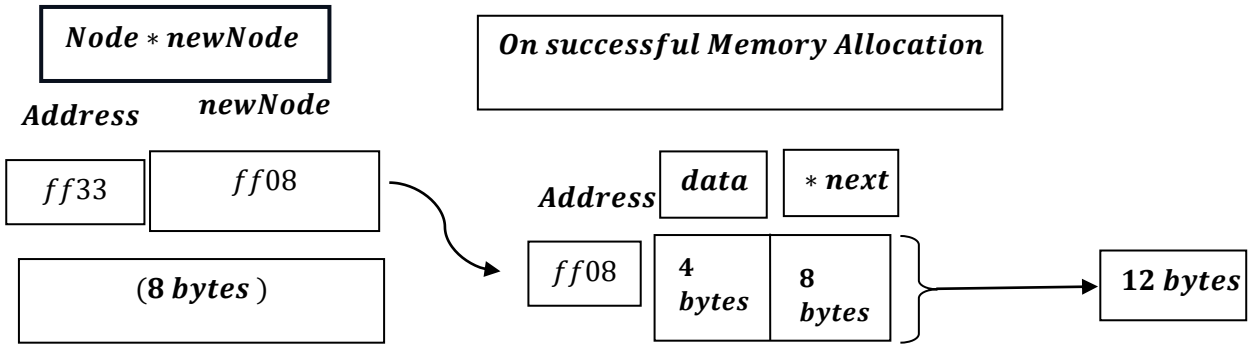
→ *Therefore , the size of the Node structure is typically 12 bytes(4 bytes for data + 8 bytes for next) .*

```
Node *newNode = (Node *)malloc(sizeof(Node));
```

What does it mean?

→ *newNode will try to allocate memory for data i.e. 4 bytes , next pointer variable i.e. typically 8 bytes, i.e. total 12 bytes, as discussed earlier , the size of a pointer (newNode) is typically 8 bytes on most modern 64 – bit systems, regardless of the size of the structure it points to.*

Say,



The above is Theoretical , But if we do a reality check:

Theoretical vs. Real-World Allocation

In a real programming environment $\begin{pmatrix}\textbf{like the C++ environments}\\ \textbf{you typically work in}\end{pmatrix}$

, the allocation might not be exactly 12 bytes because of

how CPUs access memory:

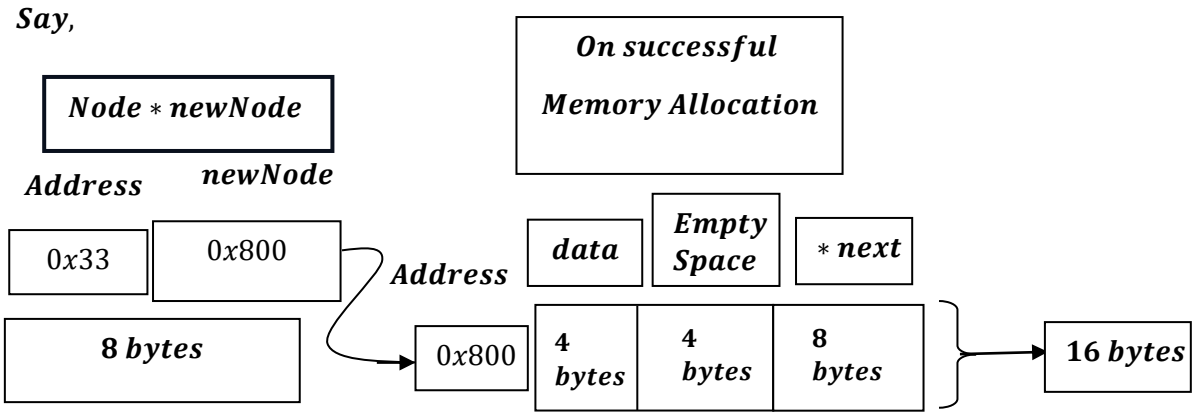
- ***Memory Alignment***: *Most* $64-$ *bit compilers align data to* $8-$ *byte boundaries to improve performance.*

- ***Padding***: *To align the* $8-$ *byte next pointer, the compiler will often add* 4 *bytes of "padding" after the* $4-$ *byte data integer.*

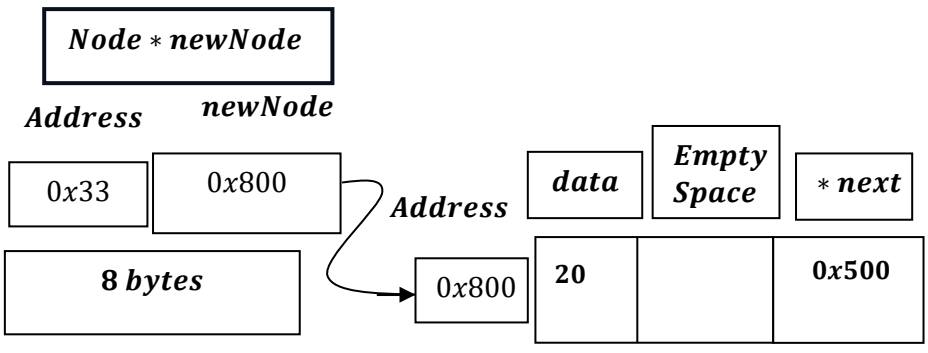
- ***Actual Size***: *Consequently, sizeof* $(Node)$ *would likely return* 16 *bytes on your system, rather than the* 12 *bytes.*

Offset	*Size*	*Content*	*Description*
0	**4** *bytes*	*data*	*The integer value* (*e.g.*, **2**).
4	**4** *bytes*	*Padding*	*Empty space added by the compiler.*
8	**8** *bytes*	*next*	*The next address of the Node in the linked list* (*e.g.*, **0x500**).

It will look like:



Like:



In a 64 − bit environment, the compiler aligns memory to the size of the largest primitive member in a structure − in this case, the 8 − byte next pointer.

Because the data integer only occupies 4 bytes, the compiler inserts empty***space (padding) to ensure the pointer starts at an address that is a multiple of 8.***

Such as:

$0x500 = 5 \times 16^2 + 0 \times 16^1 + 0 \times 16^0 = 5 \times 256 = 1280$

$1280 \div 8 = 160$ (*no remainder*)

Why so ?

As , it depends on 32 − bit system and 64 system.

In 64 − bit system , memory allocation is total : 16 bytes.

$$\begin{bmatrix} int\ variable - 4\ bytes \\ padding - 4\ bytes \\ Next\ pointer - 8\ bytes \end{bmatrix}$$

Where , as in 32 − bit system , memory allocation is total 8 bytes.

Where for pointer it is allocated − 4 bytes.

and for integer variable , memory allocated is total − 4 bytes.

Hence for 32 − bit system no, such padding is needed.
